Revised Audit Report on Q1
The Double-Octic Model, Its Singular Locus, and the Unfinished Resolution

Prepared from the Q1 polynomial, exact Singular computations,
the Rust blowup audit, and `q1_expository_report.pdf`

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Abstract

This report reconciles the newly supplied expository PDF with exact calculations for the given branch octic. The PDF and the Rust input define exactly the same polynomial $\Delta = q_4^2 - 4q_3q_5$. The PDF correctly identifies seven singular lines and an isolated point. It does not, however, correctly compute five of the seven generic transverse cA_n types. More seriously, the isolated point is already an ordinary double point of the double cover, is disjoint from every center in the advertised divisorial ledger, and lies on none of the three advertised small-resolution divisors. Consequently the claimed smooth projective model is not obtained by the construction as written. Claims found to be false or unsupported are printed in red.

Contents

1 Executive conclusion

Let Y_0 be the double cover

$$Y_0 : \quad R^2 = \Delta(a, b, c, d) \subset \mathbb{P}(1, 1, 1, 1, 4).$$

The following facts have been checked exactly over $\mathbb{Q}$.

1. The polynomial supplied to the Rust program is exactly the discriminant $q_4^2 - 4q_3q_5$ displayed in the expository PDF.
2. The branch singular locus contains the seven lines $L_1, \ldots, L_7$ listed in the PDF and the isolated point
$$P_* = [4 : -2 : 3 : 1].$$
3. The generic transverse types in line order are
$$(cA_3, cA_3, cA_3, cA_4, cA_1, cA_5, cA_1),$$
not the sequence stated in the PDF.
4. The branch Hessian at P_* has nonzero determinant, so Y_0 has an ODP above P_*. None of the listed centers meets P_*, and none of the three listed Weil divisors contains the point above P_*.

   Mistake. The assertion in Sections 5.4–5.5 of `q1_expository_report.pdf` that the advertised 36 divisorial blowups followed by the three listed Weil-divisor blowups produce a smooth threefold is false as written: the ODP above P_* is not resolved.

It follows that the correction polynomial $58p^2 + 82p$, the claimed Picard rank, Euler characteristic, rigidity, and modularity conclusion cannot be treated as proved by the provided ledger. They may ultimately be true for a corrected resolution, but a corrected global construction and new exceptional-stratum count are required.

2 Identity of the model

The PDF defines

$$q_3 = ad(b + c - d)$$

and homogeneous polynomials q_4, q_5 of degrees four and five, respectively, then sets

$$\Delta = q_4^2 - 4q_3q_5.$$

Expanding this expression in a, b, c, d gives exactly the polynomial in `q1_branch_polynomial.txt` after the identification

$$(a, b, c, d) = (x_0, x_1, x_2, x_3).$$

Thus none of the discrepancies below can be attributed to a change of coordinates or to a different octic.

The projective singular components reported by the PDF are

$L_1 : (a, b)$,	$L_2 : (a - b, c)$,	$L_3 : (a, d)$,
$L_4 : (b, c - d)$,	$L_5 : (a, c - d)$,	$L_6 : (c, d)$,
$L_7 : (a - d, b + c - d)$,		

together with

$$P_* : (c - 3d, b + 2d, a - 4d).$$

Direct substitution verifies that Δ and all four homogeneous partial derivatives vanish on each listed component.

> **Verified status.** The eight-component singular-locus decomposition in Section 4.1 of the PDF is correct. The earlier Rust report was wrong to present the seven lines as the complete initial singular locus; its projective linear fast path returned before adding the isolated point.

3 Exact transverse singularity types

3.1 Method

At a generic rational point of each line, choose an affine transverse plane with coordinates u, v. Let $f_i(u, v)$ be the restriction of Δ to that plane. The Milnor number is computed in the local ring by

$$\mu_i = \dim_{\mathbb{Q}} \frac{\mathbb{Q}[[u, v]]}{(\partial f_i/\partial u, \partial f_i/\partial v)}.$$

For these rank-one quadratic plane germs, $\mu_i = n$ is precisely the A_n index, so the double-cover family has transverse type cA_n.

The exact slices used by `q1_exact_checks.sing` are shown below.

Line	Generic point	Transverse substitution
L_1	$[0 : 0 : 1 : -1]$	$(a, b, c, d) = (u, v, 1, -1)$
L_2	$[1 : 1 : 0 : 2]$	$(a, b, c, d) = (1, 1 + u, v, 2)$
L_3	$[0 : 1 : 1 : 0]$	$(a, b, c, d) = (u, 1, 1, v)$
L_4	$[1 : 0 : -1 : -1]$	$(a, b, c, d) = (1, u, -1 + v, -1)$
L_5	$[0 : 1 : 1 : 1]$	$(a, b, c, d) = (u, 1, 1 + v, 1)$
L_6	$[1 : -1 : 0 : 0]$	$(a, b, c, d) = (1, -1, u, v)$
L_7	$[1 : 2 : -1 : 1]$	$(a, b, c, d) = (1, 2, -1 + u + v, 1 + u)$

3.2 Corrected table

Line	PDF type	Exact μ	Correct type	Generic chain length
L_1	cA_8	3	cA_3	2
L_2	cA_4	3	cA_3	2
L_3	cA_7	3	cA_3	2
L_4	cA_8	4	cA_4	2
L_5	cA_1	1	cA_1	1
L_6	cA_7	5	cA_5	3
L_7	cA_1	1	cA_1	1

For the standard local branch model $x^2 + y^{n+1}$, a crepant blowup of the double curve reduces n by two. Hence the generic chain length is $\lceil n/2 \rceil$. The corrected generic contribution is therefore

$$2 + 2 + 2 + 2 + 1 + 3 + 1 = 13.$$

Mistake. Equation (4.1) and Table 5.1 of the PDF claim 20 old-line-chain blowups based on the incorrect cA table. The generic line calculation contributes 13, not 20. Special intersection centers must then be recomputed in the corrected order; they cannot be copied unchanged from the old ledger.

4 The omitted isolated ODP

Work in the affine chart $d = 1$ and translate P_* to the origin:

$$a = 4 + u, \qquad b = -2 + v, \qquad c = 3 + w.$$

The Hessian matrix of the translated branch germ at the origin is

$$H = \begin{pmatrix} 72 & 1008 & 720 \\ 1008 & 1440 & 864 \\ 720 & 864 & 1440 \end{pmatrix}, \qquad \det(H) = -859963392 \neq 0.$$

The branch germ is therefore a three-variable node. Adding the R^2 term gives a nondegenerate quadratic form in four variables, so the point of Y_0 above P_* is an ordinary double point.

The point P_* lies on none of $L_1, \ldots, L_7$ and none of the six listed line-intersection point centers. Consequently all divisorial blowups in the PDF are isomorphisms in a neighborhood of this ODP.

At the point $[a : b : c : d : R] = [4 : -2 : 3 : 1 : 0]$, the first base equation of each advertised small-resolution divisor evaluates as follows:

Divisor	First equation	Value at P_*
$D_{ab,+}$	$b - a$	-6
$D_{b,-}$	b	-2
$D_{d,-}$	d	1

Thus none of the three divisors contains the ODP above P_*.

Mistake. The Macaulay2 appendix checks only that the three divisor ideals are contained in the double cover. It does not check that they contain every node, that their blowups resolve every node, or that the resulting threefold is smooth. Divisor containment alone is insufficient.

5 Comparison with the Rust blowup audit

5.1 What the 33-stage file actually proves

The Rust path recorded in `seven_line_stage33_raw.json` consists of 33 global curve-center stages and 105 chartwise transforms. Every recorded center has codimension two and branch multiplicity two, so every recorded stage satisfies

$$2 - 1 - \left\lfloor \frac{2}{2} \right\rfloor = 0.$$

This proves crepantness of the recorded stages, but the file was written with its terminal audit intentionally disabled. It is a checkpoint, not a resolution certificate, and its blowup order is not the 36-center order claimed by the PDF.

On charts 212 and 213, exact reduction proves that two one-dimensional components remain. For example, on chart 212 one component is

$$C_2 : \quad x_2 + x_3^2 = 0, \qquad x_0 x_3^2 - x_3 - 1 = 0.$$

Reducing the branch polynomial and all three partial derivatives modulo this ideal gives zero. The corresponding chart-213 equations are

$$x_0 x_3^2 + 1 = 0, \qquad x_2 + x_3 + 1 = 0.$$

The four bounded-grid samples previously treated as four points are paired by the overlap:

$$(0, -1, -1)_{212} = (-1, 0, -1)_{213},$$
$$(2, -1, 1)_{212} = (-1, -2, 1)_{213}.$$

Both global samples lie on C_2 and therefore are not isolated ODPs.

> **Mistake.** The earlier audit report was wrong to call the stage-33 state a "final singular locus" and wrong to report a global terminal ODP count of zero. What was proved is narrower: zero of those four chart samples are ODPs. The global terminal count is unknown, and the initial ODP above P_* was omitted from that report.

5.2 Why the two ledgers cannot yet be merged

The PDF ledger begins with incorrect generic line data and does not include transform-by- transform equations. The Rust ledger uses a different curve-first order, omitted the isolated point, and stops with nonlinear singular curves. Therefore neither existing ledger is a complete verified resolution, and their center counts cannot be combined arithmetically.

6 Status of the downstream claims

Claim	Status	Reason
Projection gives $R^2 = \Delta$	Verified	Direct algebraic identity.
Seven lines plus P_*	Verified	Exact projective Jacobian decomposition; direct substitution.
PDF transverse cA table	False	Five entries disagree with exact local Milnor numbers.
20 old line-chain blowups	False	Correct generic contribution is 13.
36-center ledger resolves the branch	Not established	The ledger is hardcoded; no successive strict-transform audit is supplied.
Only four final nodes remain	False as written	The ODP above P_* is untouched and absent from the node list.

Claim	Status	Reason
Three divisors give a smooth projective model	False as written	None of the divisors contains the ODP above P_*.
$\#Y_0(\mathbb{F}_p)$ values	Separately checkable	These count the singular double cover and do not require a resolution.
$58p^2 + 82p$ correction	Unsupported	It depends on the invalidated exceptional-stratum ledger.
$h^{1,1} = 66$, $e = 132$, rigidity	Unsupported	These depend on the same unresolved geometry.
Level-13 modularity of the claimed X	Unsupported here	The trace sequence uses the unsupported resolved counts.

The arithmetic equalities in the PDF may be internally consistent once one assumes $58p^2 + 82p$ and $h^{1,1} = 66$. Internal consistency is not a substitute for proving the geometric correction formula.

7 Work required for a valid final theorem

A corrected proof should supply all of the following.

1. Start from the corrected transverse table and recompute the ordered strict transforms of all seven lines and their special intersections.
2. Include the isolated ODP above P_* from the outset. Supply a global Weil divisor containing it and verify that its blowup gives a projective small resolution, or use another valid projective crepant construction.
3. At every divisorial stage, record the actual center ideal, branch multiplicity, branch transform, active affine charts, and exact singular locus after the transform.
4. Resolve or account for nonlinear residual curve components such as the chart-212 component C_2 in the Rust order.
5. Prove terminal smoothness by an exact Jacobian audit on every final chart.
6. Only then recompute the exceptional-stratum point-count correction, Picard rank, Euler characteristic, Hodge numbers, and Frobenius traces.

Until these steps are completed, the correct conclusion is not that Q1 has four final ODPs, but that the proposed resolution and its numerical consequences remain unproved.

8 Reproducibility files

All files used for this audit are in the `Q1/` directory.

`README.md` Directory manifest and commands for rerunning the checks.

`q1_revised_report.tex/pdf` This report and its compiled PDF.

q1_revised_findings.json Machine-readable corrected conclusions and explicit unresolved fields for the final singular locus and ODP count.

q1_expository_report.pdf The newly supplied expected-result report.

q1_branch_polynomial.txt The supplied octic.

q1_exact_checks.sing/txt Exact transverse and isolated-point checks.

q1_stage33_curve_checks.sh/txt Exact residual-curve reductions.

seven_line_initial.json Initial Rust singular-component output. It records the seven lines but omits P_*, as explained above.

seven_line_stage33_raw.json Full 33-stage raw checkpoint.

seven_line_crepant_audit.json Earlier merged audit. Its `resolution_complete=false` flag remains essential; its terminal wording is superseded by this report.

A Exact command output

The principal output of `Singular -q q1_exact_checks.sing` is

```
L1_MILNOR=3 TYPE=cA3
L2_MILNOR=3 TYPE=cA3
L3_MILNOR=3 TYPE=cA3
L4_MILNOR=4 TYPE=cA4
L5_MILNOR=1 TYPE=cA1
L6_MILNOR=5 TYPE=cA5
L7_MILNOR=1 TYPE=cA1
ISOLATED_POINT_BRANCH_VALUE=0
ISOLATED_POINT_MILNOR=1
ISOLATED_POINT_HESSIAN_DET=-859963392
ISOLATED_POINT_DOUBLE_COVER_TYPE=ODP
Dab_plus_FIRST_GENERATOR_AT_POINT=-6
Db_minus_FIRST_GENERATOR_AT_POINT=-2
Dd_minus_FIRST_GENERATOR_AT_POINT=1
```

The stage-33 curve check reports dimension one and zero remainders for the branch and all partials modulo each listed chart ideal.